HANOI UNI. OF SCI. AND TECH. **FACULTY OF MATH. AND INFORMATICS**

MIDTERM TEST, SPRING 2025 **Course: CS370 - Nat. of PL** **Duration: 60 minutes**	**Student name:** ..... **Student Id:** ..... **Order:** ... **Class Id:** .....

Evaluators' name and signature	**Proctors' name and signature**	**Total score**

Test Id: 1 **(This test consists of 15 questions)**

Note: *No document is allowed.*

Questions with one correct answers

Question 1 Strings represented by $[0-9]+$ can be

☐ ε, 10, 123, 579, 90 ☐ [0, 10, 123, 579, 90] ☐ [0], [10], [123], [579], [90] ☐ 0, 10, 123, 579, 90

Question 2 The output of a Lexical Analyzer is

☐ A sequence of lexemes ☐ A parse tree

☐ A sequence of tokens ☐ A sequence of keywords

Question 3 Which of the strings is produced by the following grammar?

$$
\begin{aligned}
&S \longrightarrow expr\\
&expr \longrightarrow expr + expr \mid expr * expr \mid (expr) \mid number\\
&number \longrightarrow number\ digit \mid digit\\
&digit \longrightarrow 0 \mid 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9
\end{aligned}
$$

☐ `2*3*4-5` ☐ `0+01*2`

☐ `00+1*2+` ☐ `x:=y+z`

Question 4 The bottom-up parsers can also be called

☐ Top-down parsers ☐ LL parsers

☐ Recursive-descent parsers ☐ Shift-reduce parsers

Question 5 In Prolog, suppose that the following clauses

$$
\begin{aligned}
&ancestor(X,Y) :- parent(X,Z),\ ancestor(Z,Y).\\
&ancestor(amy, bob).\\
&parent(eve, amy).
\end{aligned}
$$

have been entered into the database, then which of the following queries would cause the "yes" response?

☐ `?- ancestor(amy,eve).` ☐ `?- ancestor(eve,bob).`

☐ `?- ancestor(eve,amy).` ☐ `?- ancestor(bob,amy).`

Question 6 Which of the following Pascal programs causes a syntax error?

☐ `var i: Integer; begin Writeln(i:=1); end.` ☐ `var i: Integer; begin Writeln(i=1); end.`

☐ `var i: Integer; begin Writeln; i:=1.0; end.` ☐ `var i: Integer; begin Writeln; i:=1; end.`

Question 7 Which of the following parsers constructs parse trees from the root to the leaves?

☐ Bottom-up parsers ☐ Shift-reduce parsers

☐ Top-down parsers ☐ LR parsers

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Question 8 Suppose that we have the following grammar

(1) $S \longrightarrow expr$
(2) $expr \longrightarrow expr + expr \mid expr * expr \mid (expr) \mid number$
(3) $number \longrightarrow number\ digit \mid digit$
(4) $digit \longrightarrow 0 \mid 1 \mid 2 \mid 3 \mid 4 \mid 5 \mid 6 \mid 7 \mid 8 \mid 9$

Then, the order of grammar rules applied by bottom-up parsers to construct the parse tree for the string "2+3*4" will be?

☐ `4-3-2-2-4-2-2-4-3-2-2-1`
☐ `4-4-4-3-3-2-2-2-3-2-2-1`
☐ `4-3-2-4-3-2-2-4-3-2-2-1`
☐ `4-3-2-4-3-3-2-4-3-2-2-1`

Questions with multiple correct answers (Find all correct answers)

Question 9 In TinyAda, which of the following statements are legal?

☐ `integer: boolean;`
☐ `while i <= 5 do J := J + 1; end loop;`
☐ `I = I + 1;`
☐ `while i <= 5 loop J := J + 1; end while;`
☐ `type COLUMN_INDEX is range 1..10;`
☐ `while i <= 5 do J := J + 1; end while;`

Question 10 In Java, which of the following statements are legal?

☐ `System.out.println("Hello World");`
☐ `System.out.println('Hello World');`
☐ `System.out.println(Hello World);`
☐ `int v=0; if (v == 0) v=v+1;`
☐ `int v; if (v == 0) v=v+1;`
☐ `if (int v == 0) return v;`

Question 11 In C#, which of the following statements are legal?

☐ `if (int v == 0) return v;`
☐ `Console.out.println("Hello World");`
☐ `Console.out.println('Hello World');`
☐ `int v=0; if (v == 0) v=v+1;`
☐ `Console.WriteLine("Hello World");`
☐ `int v; if (v == 0) v=v+1;`

Question 12 In Python, which of the following statements are legal?

☐ `if (v == 0) v=v+1;`
☐ `print ("Hello World");`
☐ `sum := float(1) + float(2);`
☐ `println('Hello World');`
☐ `Console.out.println("Hello World");`
☐ `print ('Hello World');`

Fill-the-gap questions

Question 13 Languages represented by regular expressions are called

.....

Question 14 Identify tokens in the following C declaration: `const char *n_plain, *n_keytext;`

.....

Question 15 Explain in detail the meaning of the following code in C++

```
enum Disc {IsInt,IsReal};
struct IntOrReal {
  enum Disc which;
  union {
    int i;
    double r;
  };
};
```

.....

.....